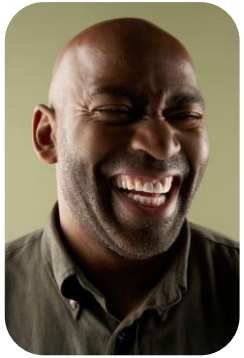
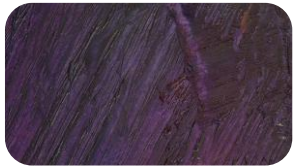
Presentation

Olist
Brazilian E-Commerce





AGENDA

- Objectives
- Data Workflow Overview
- Data Ingestion
- Data Warehouse Design
- Load - dbt
- Data Monitoring
- Analytics & Insights
- Risk & Mitigations

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MEET THE TEAM

Wong Huey Shyan

Man Wai Kay

Isaac Chin

Jaskson Wong

Lily Seah

Phoa Kim Meng

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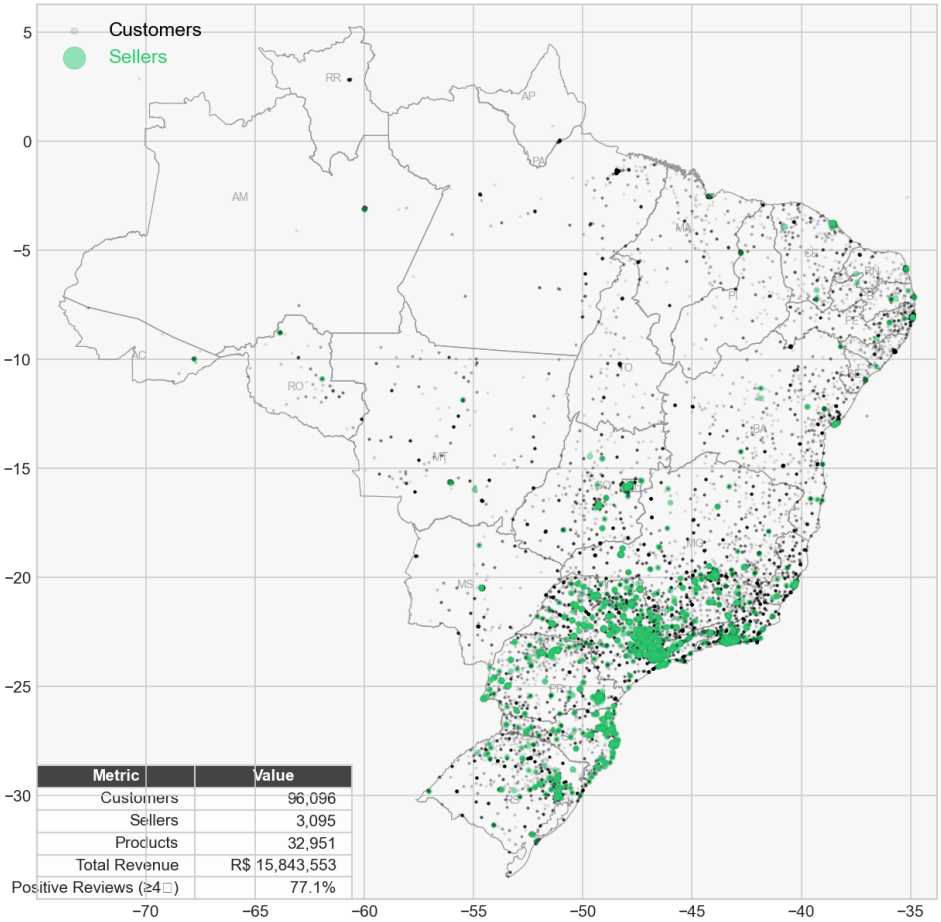
Objectives

To share our marketplace operations, including:

- End-to-end data pipeline solution
- Key findings on business analytics
- Business recommendations



Metric	Value
Customers	96,096
Sellers	3,095
Products	32,951
Total Revenue	R$ 15,843,553
Positive Reviews (≥4□)	77.1%

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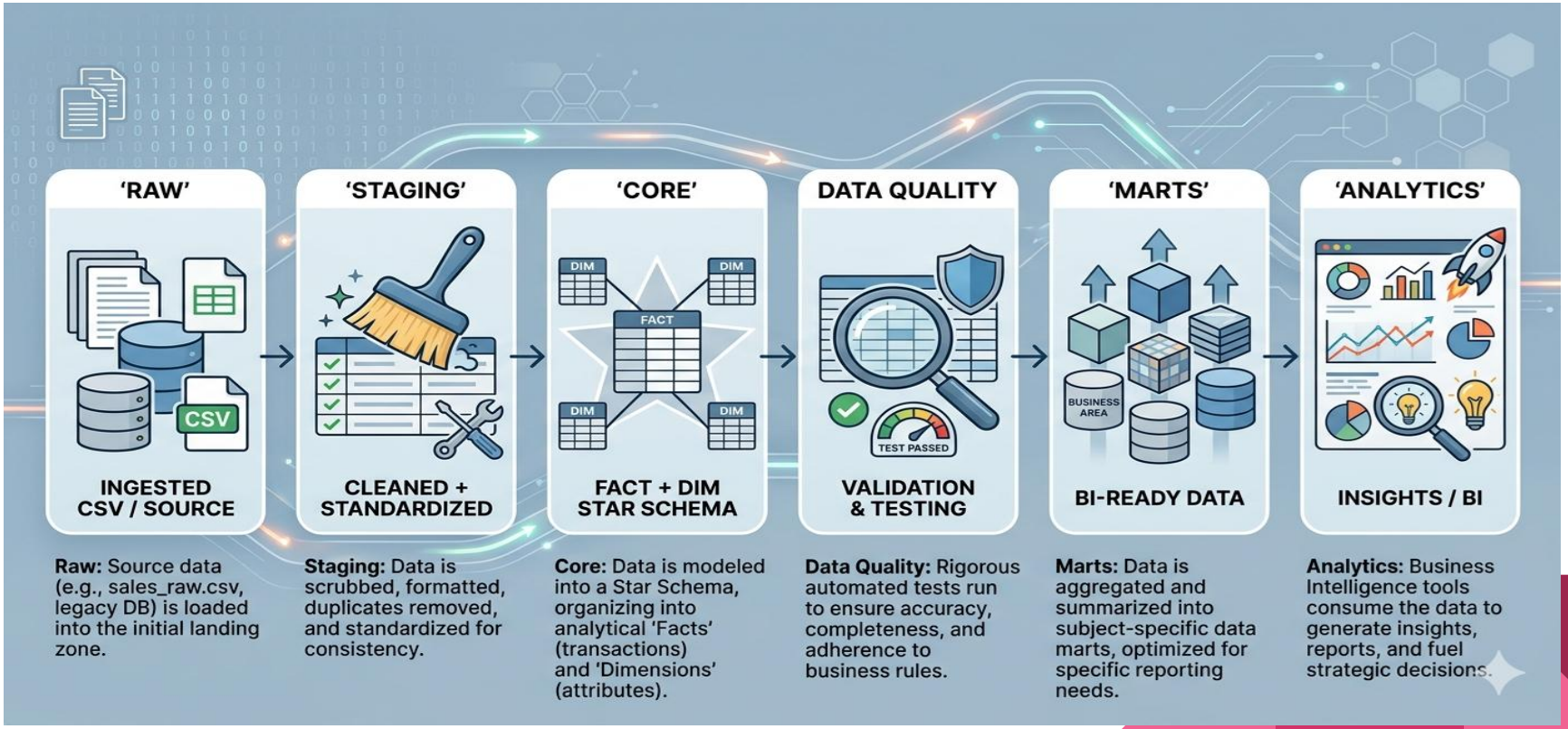
Data Workflow Overview



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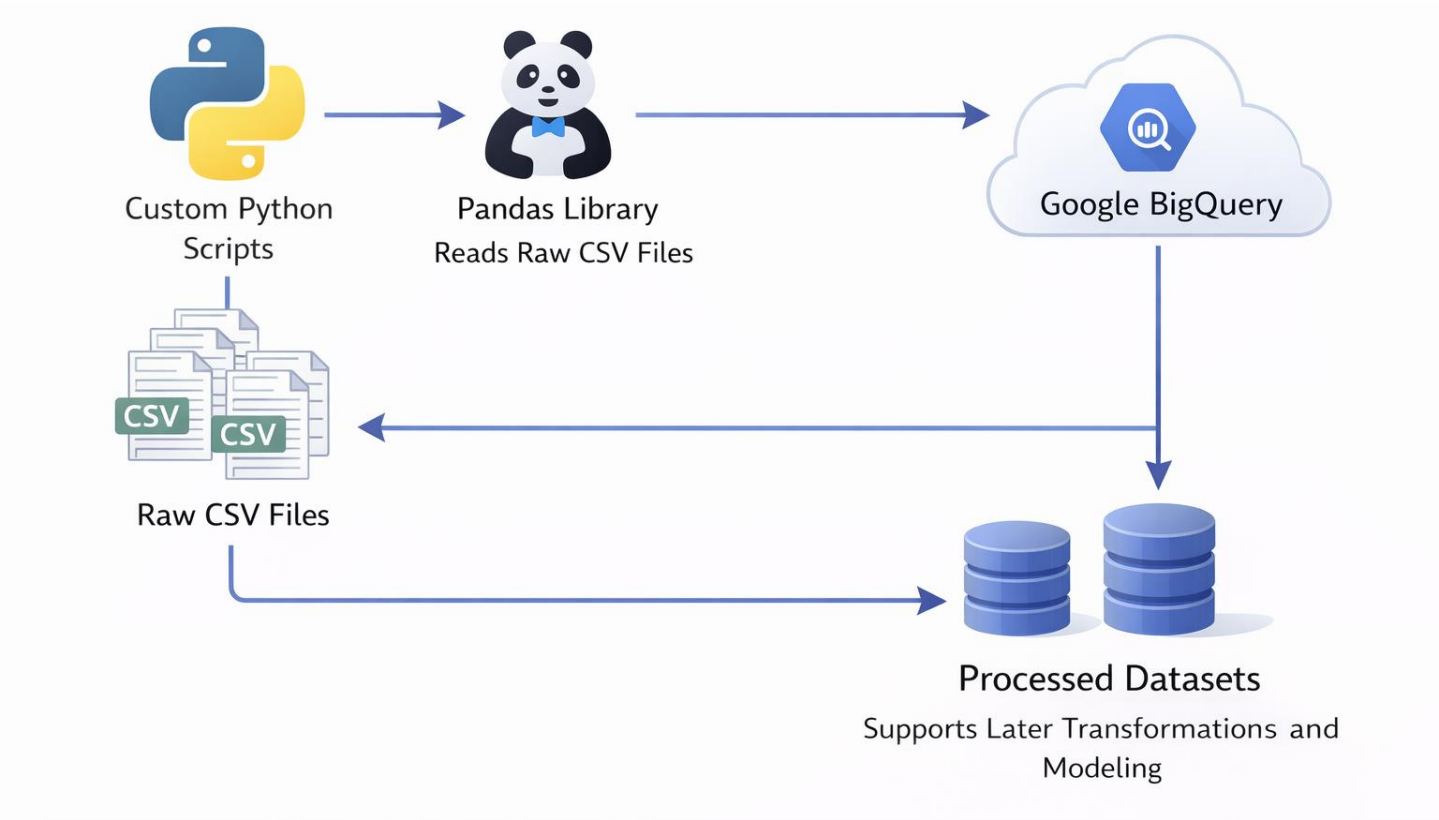
Data Workflow Overview



Data Ingestion

[01] THE EXTRACTION ENGINE: PYTHON & PANDAS

AUTOMATION FIRST

> Eliminating manual uploads for daily high-volume data.

> Scheduled Python scripts act as the primary ingestion engine.

> Scalable handling of CSV formats across various sources.

PANDAS LIBRARY

> Robust parsing and pre-processing capabilities.

> Efficiently handles large datasets into DataFrames.

```
import pandas as pd
df = pd.read_csv('orders.csv')
df.rename(columns=...)
```

INITIAL TASKS

> **Column Renaming:** Standardizing source headers.

> **Type Inference:** Ensuring correct data formats.

> **Cleaning:** Stripping whitespace and handling nulls.

DATA SOURCES: olist_orders_dataset.csv, olist_customers_dataset.csv

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Data Ingestion

[02] THE UNIFIED INGESTION PATTERN

READ → VALIDATE → STANDARDIZE → LOAD

DESIGN PRINCIPLE: CONSISTENCY

> Shared utility functions across all ingestion scripts.

> Standardized column naming conventions at load time.

RELIABILITY & AUDITABILITY

> Schema validation catches structural issues early.

> Repeatable process eliminates human error.

Data Ingestion

[03] STORAGE: GOOGLE BIGQUERY & THE RAW LAYER

CLOUD DATA WAREHOUSE

> **Serverless Scale:** Handling petabyte-scale storage without infrastructure management.

> **Direct Upload:** Processed data lands directly from Python scripts into BigQuery tables.

```
[ ARCHITECTURE ]
Python -> BigQuery API -> Raw Dataset
```

THE RAW LAYER STRATEGY

> **Immutable Record:** Mirroring source files exactly with zero business logic applied.

> **Single Source of Truth:** Essential for full reprocessing if downstream logic changes.

> **Auditability:** Preserving data in its original state for compliance and debugging.

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Data Ingestion

[04] CONCLUSION: FROM RAW DATA TO INSIGHTS

THE ANALYTICAL BACKBONE

> Ingestion acts as the foundation for all downstream modeling.

> Automating the flow unlocks data value at scale.

> Reliable, auditable, and repeatable architecture.

READY FOR TRANSFORMATION

FACT TABLE

fact_order_item

↓ JOIN

DIMENSION TABLE

dim_product, dim_customer

[UNLOCKING E-COMMERCE VALUE THROUGH AUTOMATED PIPELINES]

Data Warehouse Design

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Load - dbt

dbt acts as the **final delivery mechanism** – taking already-transformed data and materializing it into structured, tested, documented tables in the warehouse.

Considerations:

- Dependency Management (DAG) – ensure that tables are loaded in the correct order
- Load-time testing
- Do not require real time streaming

Tool	Phase	What happens
	Extract	Pull raw data from source
	Transform	Transform data outside the warehouse
dbt	**Load**	**Write final, clean data into target warehouse tables**

Data Monitoring

Where Monitoring Helps — and Where It Doesn't

Key Points:

Strengths

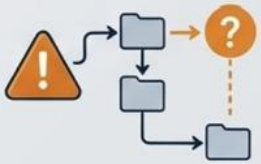
- Detects unknown issues
- Reduces manual testing
- Improves reliability quickly

Limitations

- Usually covers only selected tables
- Alerts lack full lineage context
- Doesn't fully solve root-cause analysis

Limitations

- Usually covers only selected tables



- Alerts lack full lineage context

- Doesn't fully solve root-cause analysis

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Data Monitoring

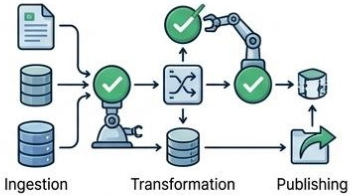
Why Use Great Expectations for Data Monitoring?

Turns data quality into code

- Declarative, version-controlled expectations for consistent monitoring.

Automates validation

- Runs checks during ingestion, transformation, and before publishing.

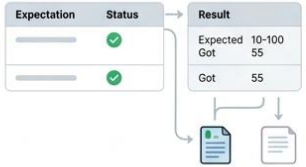


Produces clear reports

- Human-readable Data Docs for fast debugging and stakeholder visibility.

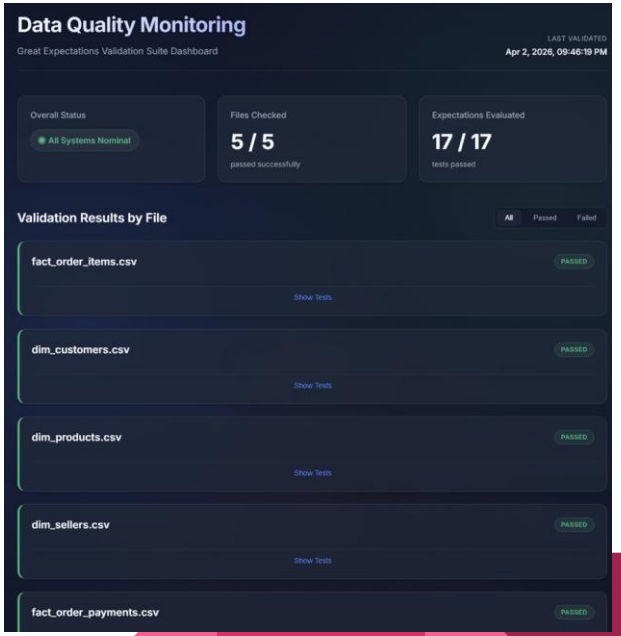
4 Fits modern data stacks

- Integrates with dbt, Airflow, Dagster, Spark, BigQuery, Snowflake.



5 Open-source + customizable

- No vendor lock-in; supports custom rules and domain-specific checks.



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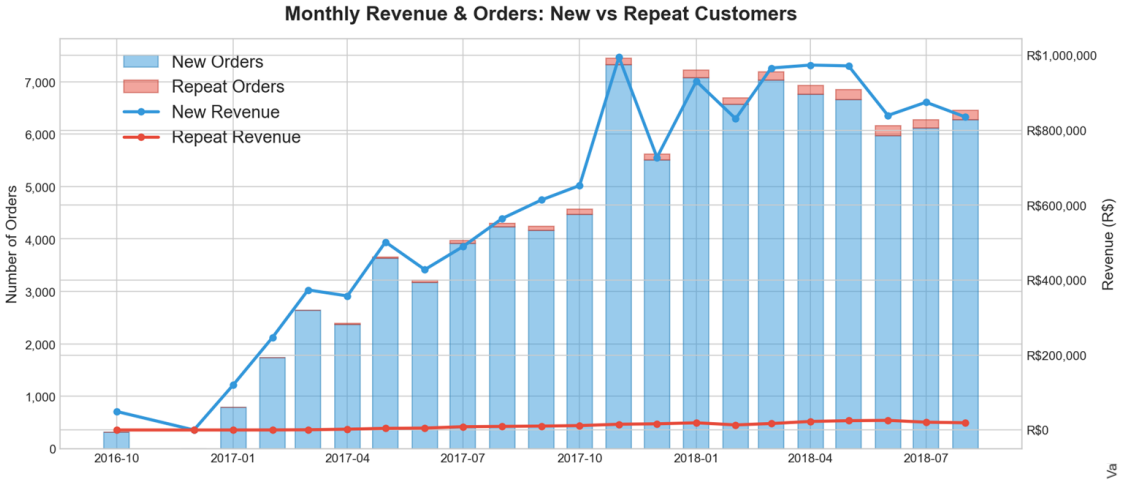
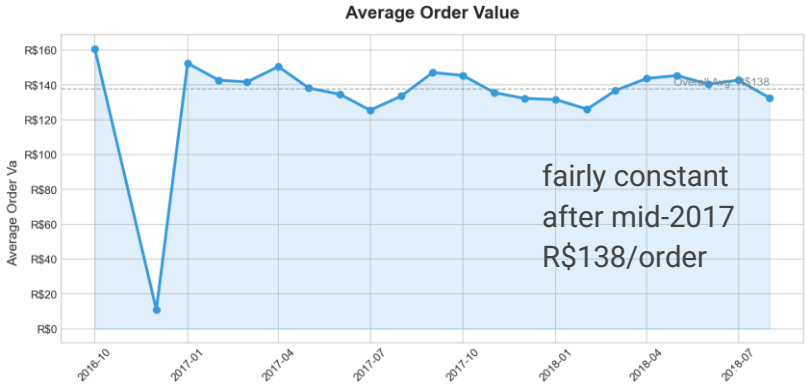
Analytics & Insights

Rapid & Sustained Acquisition Scale



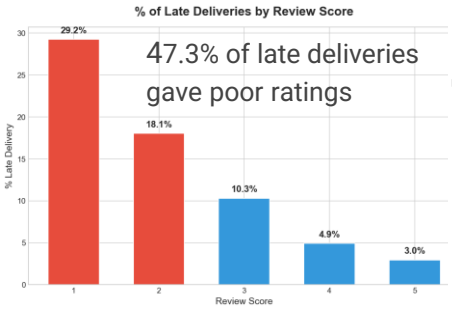
- Revenue growth 14.2% m-o-m
- Order growth 15.8% m-o-m
- Stable AOV, top 10 products account for 62.4% of rev

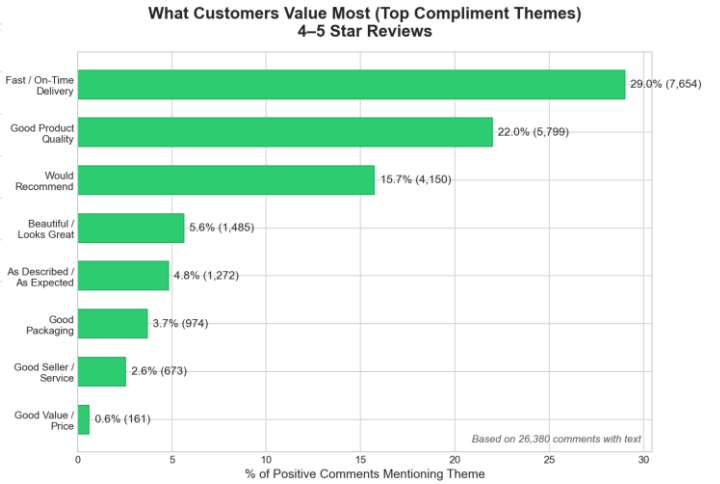
Opportunity: Acquired customers left the ecosystem

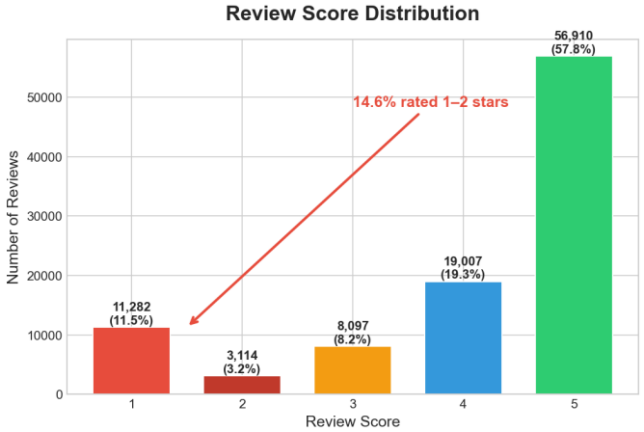


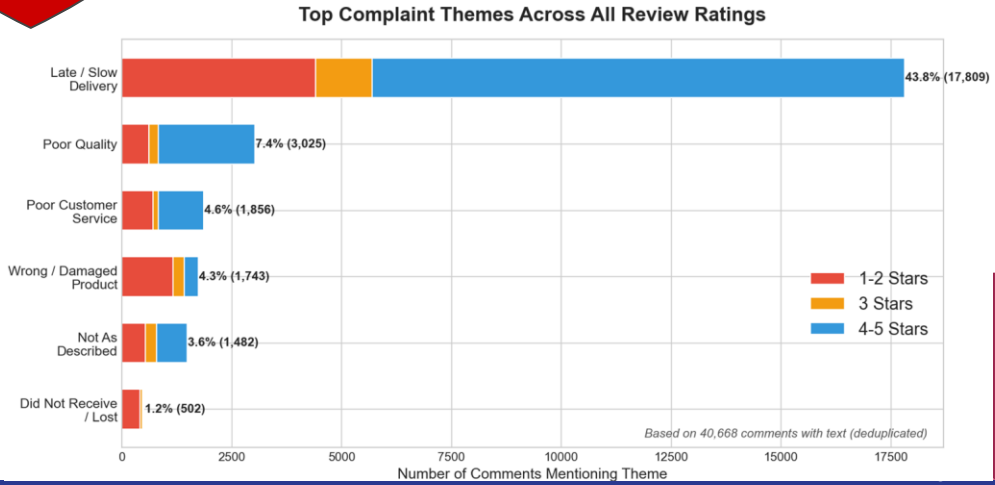
Analytics & Insights

Customer Sentiments





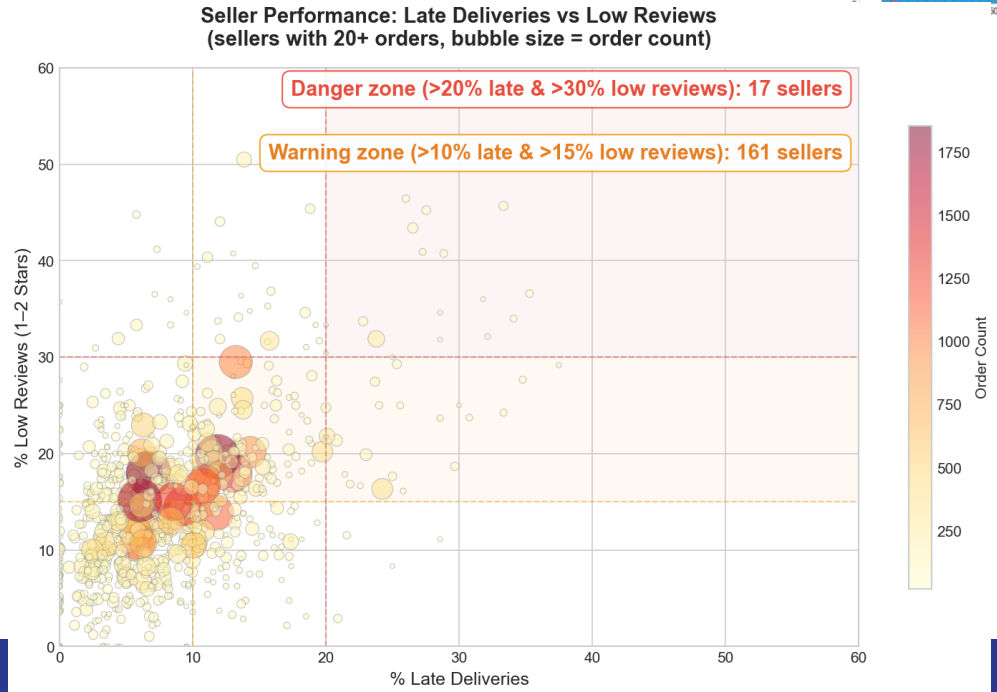




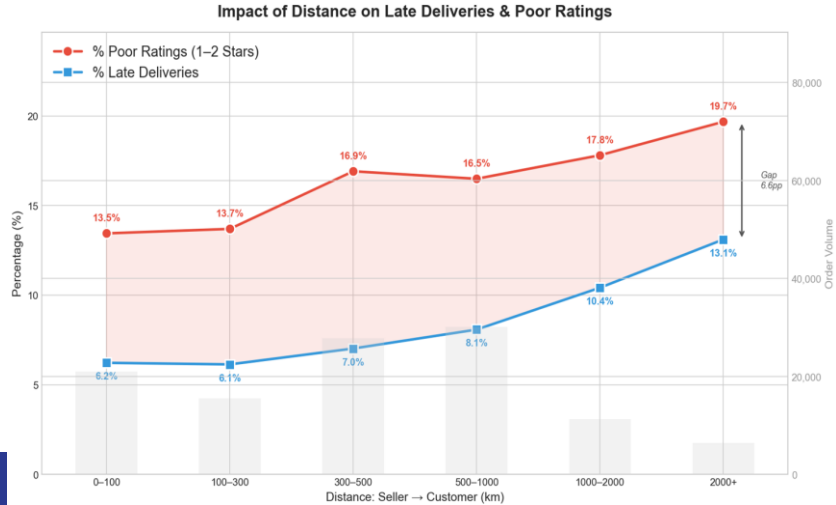
- Review: 77.1% 4-5 stars, 14.6% 1-2 stars
- Delivery: 93.2% on time, 6.8% late
- 1-2 stars: 53.4% late, 22.3% did not receive

Analytics & Insights

Almost 1 in 4 orders by sellers in warning & danger zones



Expansion Risk: chance of late delivery doubles with distance



Risks & Mitigations

Data Workflow Risks & Mitigations

Key Risks

- Data completeness issues
- Schema drift during ingestion
- Scaling challenges as data volume grows

Mitigations

- Automated quality checks
- Version-controlled transformations
- Cloud warehouse elasticity

Risks & Mitigations

Analytics Risks & Mitigations

Key Risks

- Customer churn despite high acquisition
- Delivery delays impacting ratings
- Geographic expansion increases late deliveries
- Seller quality concentration (top 20% dominate)

Mitigations

- Targeted retention programs
- Logistics optimization & SLA monitoring
- Distance-aware delivery planning
- Seller performance scoring & early-warning flags

Risks & Mitigations

Integrated Risk Strategy

Combined Mitigation Framework

- Strengthen data reliability
- Improve operational predictability
- Enhance customer experience
- Reduce seller-driven variability

THANK YOU!